

YUNYONG KO

CONTACT INFORMATION	Building 310, Room 744 84, Heukseok-ro, Dongjak-gu, Seoul, 06974, Republic of Korea	Tel: + 82 - 2 - 820 - 5507 Email: yyko@cau.ac.kr Homepage: https://yy-ko.github.io
RESEARCH INTERESTS	My research interests mainly lie in data mining and machine learning on a various types of data (e.g., graph, hypergraph, text, image, and time-series), with a special focus on exploring knowledge for real-world applications.	
EDUCATION	Hanyang University , Seoul, Korea <i>Ph.D. in Computer Science</i> - Thesis: Effective Approaches to Distributed Deep Learning: Methods, Analyses, and Evaluation - Advisor: Prof. Sang-Wook Kim Hanyang University , Seoul, Korea <i>B.S. in Computer Science</i>	Sep. 2013 – Aug. 2021 Mar. 2009 – Aug. 2013
POSITIONS	Chung-Ang University , Seoul, Korea <i>Assistant Professor, School of Computer Science and Engineering</i> University of Illionois at Urbana-Champaign , Urbana, IL, USA <i>Postdoctoral Researcher, Department of Computer Science</i> - Topic: Large-Scale Hypergraph Learning for Real-World Applications - Advisor: Prof. Hanghang Tong Hanyang University , Seoul, Korea <i>Postdoctoral Researcher, Department of Computer Science</i> - Topic: Optimization Technique for Large-Batch DNN Training - Advisor: Prof. Sang-Wook Kim The Pennsylvania State University , University Park, PA, USA <i>Visiting Scholar, College of Information Sciences and Technology</i> - Topic: Data Parallelism Approach for Distributed Deep Learning - Advisor: Prof. Dongwon Lee	Mar. 2024 – Present May. 2022 – Feb. 2024 Sep. 2021 – April. 2022 Oct. 2019 – Feb. 2020
AWARDS & HONORS	Received the Scholarship and Teaching for Engineering Postdocs (STEP) - Grainger College of Engineering (GCOE), University of Illinois at Urbana-Champaign Selected as One of the Best-Ranked Papers - IEEE International Conference on Data Mining (IEEE ICDM) Selected as One of the Spotlight Presentations - ACM International Conference on Information and Knowledge Management (ACM CIKM) Received the Outstanding Ph.D. Dissertation Award - Research Institute of Industrial Science, Hanyang University Received the ACM SIGAPP Student Travel Award - ACM Symposium on Applied Computing (ACM SAC) Awarded the NAVER Ph.D. Fellowship - Naver Corporation Domestic Conference/Journal Awards Best Paper Awards: KIPS Spring Confernece (2021, 2023), Journal of KIISE (2017) Best Presentation Award: KIISE KCC (2017)	2023 2021 2021 2021 2019 2017

- [P3] Heterophily-Aware Adaptive Knowledge Distillation for Hypergraph Neural Networks
Joohee Cho, David Yoon Suk Kang, and Yunyong Ko
arXiv: 2606.08978
- [P2] PROBE^{Web}: An Interactive System for Probing Evaluation Landscapes of KGC Models
SooHo Moon and Yunyong Ko
arXiv: 2606.08926
- [P1] Generalized Rank-based Evaluation for Knowledge Graph Completion: Perspectives, Framework, and Analyses
SooHo Moon, Jian Kang, and Yunyong Ko
arXiv: 2606.08921

International Conference and Journal Papers (* indicates equal contributions)

- [20] How Sharp and Bias-Robust is a Model? Dual Evaluation Perspectives on Knowledge Graph Completion
SooHo Moon and Yunyong Ko
WSDM'26-Short – *ACM International Conference on Web Search and Data Mining*
- [19] Accelerating Storage-Based Training for Graph Neural Networks
Myung-Hwan Jang, Jeong-Min Park, Yunyong Ko, and Sang-Wook Kim
KDD'26 – *ACM SIGKDD Conference on Knowledge Discovery and Data Mining*
- [18] Is This News Still Interesting to You? Lifetime-Aware Interest Matching for News Recommendation
Seongeun Ryu, Yunyong Ko, and Sang-Wook Kim
CIKM'25 – *ACM International Conference on Information and Knowledge Management*
- [17] Learning Short-Term and Long-Term Patterns of High-Order Dynamics in Real-World Networks
Yunyong Ko, Daeun Lee, Song Kyung You, and Sang-Woo Kim
CIKM'25-Short – *ACM International Conference on Information and Knowledge Management*
- [16] CROWN: A Novel Approach to Comprehending Users' Preferences for Accurate Personalized News Recommendation
Yunyong Ko, Seongeun Ryu, and Sang-Wook Kim
WWW'25 – *ACM Web Conference*
Selected as an Oral Presentation of WWW 2025
- [15] HyGEN: Regularizing Negative Hyperedge Generation for Accurate Hyperedge Prediction
Song Kyung Yu, Da Eun Lee, Yunyong Ko, and Sang-Wook Kim
WWW'25-Short – *ACM Web Conference*
- [14] Enhancing Hyperedge Prediction with Context-Aware Self-Supervised Learning
Yunyong Ko, Hanghang Tong, and Sang-Wook Kim
TKDE – *IEEE Transactions on Knowledge and Data Engineering*
SCIE Journal (Q1: IF Top 1.8%)
- [13] HearHere: Mitigating Echo Chambers in News Consumption through an AI-based Web System
Youngseung Jeon, Jaehoon Kim, Sohyun Park, Yunyong Ko, Seongeun Ryu, Sang-Wook Kim, and Kyungsik Han
CSCW'24 – *ACM Conference on Computer-Supported Cooperative Work and Social Computing*
- [12] SAGE: A Storage-Based Approach for Scalable and Efficient Sparse Generalized Matrix-Matrix Multiplication
{Myung-Hwan Jang*, Yunyong Ko*}, Hyuck-Moo Gwon, Ik-Hyeon Jo, Yongjun Park, and Sang-Wook Kim
CIKM'23 – *ACM International Conference on Information and Knowledge Management*
- [11] KHAN: Knowledge-Aware Hierarchical Attention Networks for Accurate Political Stance Prediction

- Yunyong Ko, Seongeun Ryu, Soeun Han, Youngseung Jeon, Jaehoon Kim, Sohyun Park, Kyungsik Han, Hanghang Tong and Sang-Wook Kim
WWW'23 – *ACM Web Conference*
- [10] RealGraph^{GPU}: A High-Performance GPU-Based Graph Engine Toward Large-Scale Real-World Network Analysis
 Myung-Hwan Jang, Yunyong Ko, Dongkyu Jeong, Jeong-Min Park, and Sang-Wook Kim
CIKM'22-Short – *ACM International Conference on Information and Knowledge Management*
- [9] Not All Layers Are Equal: A Layer-Wise Adaptive Approach Toward Large-Scale DNN Training
 Yunyong Ko, Dongwon Lee, and Sang-Wook Kim
WWW'22 – *ACM Web Conference*
- [8] D-FEND: A Diffusion-Based Fake News Detection Framework for News Articles Related to COVID-19
 Soeun Han, Yunyong Ko, Yusim Kim, Heejin Park, Seongsu Oh, and Sang-Wook Kim
SAC'22 – *ACM Symposium on Applied Computing*
- [7] MASCOT: A Quantization Framework for Efficient Matrix Factorization in Recommender Systems {Yunyong Ko*, Jae-Seo Yu*}, Hong-Kyun Bae, Yongjun Park, Dongwon Lee, and Sang-Wook Kim
ICDM'21 – *IEEE International Conference on Data Mining*
Selected as One of the Best-ranked Papers of ICDM 2021 for Fast-track Journal Invitation
- [6] ALADDIN: Asymmetric Centralized Training for Distributed Deep Learning
 Yunyong Ko, Kibong Choi, Hyunseung Jeon, Dongwon Lee, and Sang-Wook Kim
CIKM'21 – *ACM International Conference on Information and Knowledge Management*
Selected as a Spotlight Presentation of CIKM 2021
- [5] An In-Depth Analysis of Distributed Training of Deep Neural Networks
 Yunyong Ko, Kibong Choi, Jiwon Seo, and Sang-Wook Kim
IPDPS'21 – *IEEE International Parallel and Distributed Processing Symposium*
- [4] Influence Maximization for Effective Advertisement in Social Networks: Problem, Solution, and Evaluation
 Suk-Jin Hong, Yunyong Ko, Moon-Jeung Joe, and Sang-Wook Kim
SAC'19 – *ACM Symposium on Applied Computing*
- [3] Efficient and Effective Influence Maximization in Social Networks: A Hybrid-Approach
 Yunyong Ko, Kyung-Jae Cho, and Sang-Wook Kim
Information Sciences
SCIE Journal (Q1: IF Top 5%)
- [2] Influence Maximization in Social Networks: A Target-Oriented Estimation
 Yunyong Ko, Dong-Kyu Chae, and Sang-Wook Kim
Journal of Information Science
SCIE Journal (Q2)
- [1] Accurate Path-Based Influence Maximization in Social Networks
 Yunyong Ko, Dong-Kyu Chae, and Sang-Wook Kim
WWW'16-Short – *ACM Web Conference*

INVITED TALKS

Generalized Rank-based Evaluation for Knowledge Graph Completion

- Invited Talk @ Chungbuk National University (CBNU), Jan. 2026
- Invited Talk @ Ulsan National Institute of Science and Technology (UNIST), Jan. 2026

Efficient and Scalable Approach for Graph Machine Learning

- Invited Talk @ Korea-Japan Database (KJDB) Workshop, Nov. 2024

Knowledge Discovery from Real-world Networks: Dilemma, Methods, and Applications

- Invited Talk @ Data Intelligence Workshop, Jul. 2024
- Invited Talk @ New Faculty Session of KCC 2024, June. 2024
- Invited Talk @ Hanyang University (HYU), Feb. 2024

KHAN: Knowledge-Aware Hierarchical Attention Networks for Accurate Political Stance Prediction

- Invited Talk @ Electronic & Information Research Information Center (EIRIC), Sep. 2023

Not All Layers Are Equal: A Layer-Wise Approach Towards Large-Scale DNN Training

- Poster Session @ Hyundai Vision Conference (HVC), Aug. 2023
- Invited Talk @ METU-HYU Joint Workshop, Dec. 2022

Basic Concept of Distributed Deep Learning with PyTorch Tutorials

- Invited Talk @ Medical AI Korea, Oct. 2021

Efficient and Effective Influence Maximization in Social Networks: A Hybrid-Approach

- Invited Talk @ Waseda-UMS-Hanyang-UKM (WUHU) Joint Workshop, Dec. 2018
- Invited Talk @ NAVER Corp., Feb. 2018

TEACHING

Graduate School

- CAU Advanced Database (01653) Spring 2026
- CAU Data Mining (48768) Fall 2025-2026

Undergraduate School

- CAU Data Science (59193) Spring 2026
- CAU Capstone Design (56119, 56120) Fall, Spring 2025
- CAU Database (59190) Fall 2025
- CAU Algorithm (13601) Spring 2024 - 2025
- CAU Artificial Intelligence (17182) Fall 2024
- CAU Data Structure (40989) Spring 2024

PROFESSIONAL SERVICES

Track Co-Chair

- ACM Symposium on Applied Computing (SAC) 2023 - 2025

Program Committee Member (or Conference Reviewer)

- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD) 2021 - 2022, 2024 - 2026
- ACM Web Conference (WWW) 2023 - 2025
- ACM International Conference on Information and Knowledge Management (CIKM) 2025, 2026
- IEEE International Conference on Data Mining (ICDM) 2022, 2023
- AAAI International Conference on Artificial Intelligence (AAAI) 2021
- ACM Symposium on Applied Computing (SAC) 2022 - 2025
- International Conference on Database Systems for Advanced Applications (DASFAA) 2026
- IEEE International Conference on Data Science and Advanced Analytics (DSAA) 2025

Journal Topic Editor

- Frontiers in Big Data (Data Science Section)
Special Issue: Graph-based Retrieval-Augmented Generation (RAG) Systems 2025

Journal Reviewer

- ACM Computing Surveys (CSUR) 2024, 2026
- ACM Transactions on Knowledge Discovery from Data (TKDD) 2025

- IEEE Transactions on Knowledge and Data Engineering (**TKDE**) 2026
- IEEE Transactions on Neural Networks and Learning Systems (**TNNLS**) 2023, 2025
- IEEE Transactions on Network Science and Engineering (**TNSE**) 2024
- Knowledge and Information Systems (**KAIS**) 2025
- Journal of Supercomputing 2023

PATENTS

Granted Patents

- Layer-Wise Learning Rate Scheduling for Large-Scale Data Parallel Based Deep Learning
Registration Number: **KR10-2890162** Dec. 2025
- Asymmetric Centralized Training for Distributed Deep Learning
Registration Number: **KR10-2555268** Jul. 2023
- Multi-State Diffusion Model Using Interest, Intimacy, and Share Tendency
Registration Number: **KR10-2332348** Nov. 2021
- Accurate Ad-Effect Estimation Method based on Relevance between User and Item
Registration Number: **KR10-2144122** Aug. 2020
- Hybrid Approach for Influence Maximization in Social Networks
Registration Number: **KR10-1810864** Dec. 2017

Filed Patents

- Personalized News Recommendation Technique using Category-guided Intent Disentanglement and Consistency-based News Representation
Application Number: **KR10-2023-0196006** Dec 2023
- Political Stance Prediction Method and System using Political Knowledge Graphs and Hierarchical Attention Networks
Application Number: **KR10-2023-0059346** May 2023